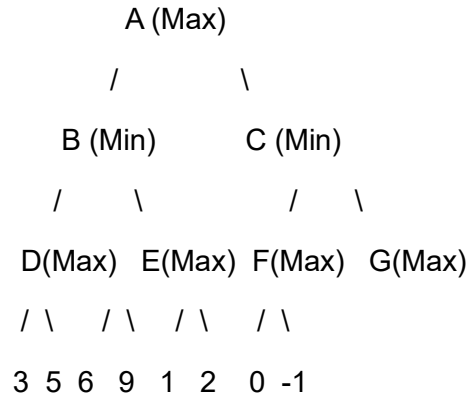




AI LAB 8 Tasks

Task 1: Minimax on Custom Game Tree

Given the following game tree:



- Apply the Minimax algorithm.
- Compute values for all internal nodes (D, E, F, G, B, C, A).
- Identify:
 - Optimal value at root node A
 - Best move for Max player
- Show step-by-step backtracking process.

Task 2: Alpha-Beta Pruning on Game Tree

Use the same tree from Task 1.

- Apply Alpha-Beta pruning step-by-step
- Track:
 - Alpha (α) and Beta (β) values at each node
- Identify:
 - Pruned branches
 - Number of nodes evaluated vs Minimax
- Compare efficiency:
 - Nodes explored (Minimax vs Alpha-Beta)



AI LAB 8 Tasks

Task 3: Minimax with Depth Limit & Heuristic

Design a grid-based game (1D or 2D) where:

- Agent (Max) tries to reach a goal (+10 reward)
- Opponent (Min) tries to block (-10 penalty)

Task:

- Implement Minimax with:
 - Depth limit = 3
 - Custom heuristic function (e.g., distance to goal)
- Simulate at least 3 game states
- Print:
 - Chosen move at each step
 - Heuristic values at leaf nodes
- Compare:
 - Behavior with depth=2 vs depth=3

Task 4: Alpha-Beta in Tic-Tac-Toe AI

Implement a **Tic-Tac-Toe AI Agent** where:

- AI = Max player (X)
- Human = Min player (O)

Task:

1. Implement:
 - a. Alpha-Beta pruning algorithm
 - b. Game board (3×3)
2. AI should:
 - a. Always play optimally
 - b. Never lose (win or draw guaranteed)
3. Display:
 - a. Game tree depth explored
 - b. Moves pruned during execution